

LIST OF MATRIX OPERATIONS TO KNOW

1. Matrix dimension $n \times m$ refers to $n = \# \text{rows}$, $m = \# \text{columns}$. If $m=n$, the matrix is a *square matrix*.
2. Matrix notation: $\mathbf{A} = \begin{pmatrix} a_{11} & \cdot & a_{1m} \\ \cdot & \cdot & \cdot \\ a_{n1} & \cdot & a_{nm} \end{pmatrix}$
3. Addition of matrices--- straightforward-- add corresponding elements. Matrices must be of same dimension.
4. Multiplication of a matrix by a constant-- multiply each element by the constant.
5. Multiplication of two matrices---row 1 of the first matrix multiplies corresponding elements of column 1 of the second matrix, then these products are summed up to form *product element* (1,1) of the new matrix, row 1 with column 2 forms *product element* (1,2) etc. When multiplying $\mathbf{A}$ by $\mathbf{B}$ to form $\mathbf{AB}$, # columns in $\mathbf{A}$ has to equal #rows in $\mathbf{B}$.
6. Identity matrix $\mathbf{I}$ has 1's on the diagonal and 0's elsewhere. It is a square matrix of any dimension. When multiplying $\mathbf{A}$ by $\mathbf{I}$, one gets back $\mathbf{A}$. In fact $\mathbf{AI} = \mathbf{A}$ and $\mathbf{IA} = \mathbf{A}$, as long as the matrices can be multiplied.
7. Transpose of $\mathbf{A}$ denoted by $\mathbf{A}'$. The transpose is obtained by flipping rows and columns. Note that $\mathbf{AA}'$ is a square matrix, as is $\mathbf{A}'\mathbf{A}$. However, the two do not usually equal each other. In fact, if $\mathbf{A}$ is $n \times m$, then $\mathbf{AA}'$ is $n \times n$, but $\mathbf{A}'\mathbf{A}$ is $m \times m$.
8. Symmetric matrix-- $\mathbf{A}$ is symmetric if $\mathbf{A} = \mathbf{A}'$. Symmetric matrices are important as they will often be encountered in statistics. $\mathbf{AA}'$ as well as $\mathbf{A}'\mathbf{A}$ are symmetric matrices.
9. The transpose of a product. It can be demonstrated that $(\mathbf{AB})' = \mathbf{B}'\mathbf{A}'$. We will need this when showing how the standard error of the estimators of the regression parameters are obtained.
10. Cofactor of a matrix element a_{ij} -- the determinant of the smaller matrix obtained when row i and column j are deleted., multiplied by $(-1)^{i+j}$.
11. Determinant: The determinant of any $n \times n$ matrix is $\sum_{i=1}^n a_{i1} (-1)^{i+1} A_{i1}$ where A_{i1} is the determinant of the matrix obtained when 1st column and the i th row are deleted, and $(-1)^{i+1} A_{i1}$ is the cofactor of element a_{i1} . Note that the

determinant of a 2×2 matrix $\begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$ (which by the cofactor formula is derived as $(-1)^2 ad + (-1)^3 cb$). In examples, we will only use the determinant of a 2×2 matrix.

12. Inverse $\mathbf{A}^{-1}$. Defined by $\mathbf{A}\mathbf{A}^{-1} = \mathbf{A}^{-1}\mathbf{A} = \mathbf{I}$. A matrix is invertible if it is square and has non-zero determinant. A non-invertible square matrix is sometimes referred to as *singular* (and an invertible one as *non-singular*). $\mathbf{A}^{-1}$ is obtained by $\frac{1}{\text{determinant}}$ (cofactor matrix)'. It is handy to know that

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \frac{1}{ad-bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$

13. Diagonal matrices have especially simple properties. Obtaining the inverse involves inverting each of the diagonal elements (i.e. $1/a_{11}$ etc.). When a diagonal matrix has the same entry along its entire diagonal, multiplying another matrix by it has the same effect as multiplying by that constant. For example:

$$\begin{pmatrix} a & 0 & 0 \\ 0 & a & 0 \\ 0 & 0 & a \end{pmatrix} \mathbf{B} = a\mathbf{B}$$

14. When there is a string of matrices multiplied by each other, a constant multiplier anywhere in the string can be moved up front. For example:

$$\mathbf{A} \begin{pmatrix} k & k \\ k & k \end{pmatrix} \mathbf{B} = k\mathbf{A} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \mathbf{B}$$

and

$$\mathbf{A} \begin{pmatrix} k & 0 \\ 0 & k \end{pmatrix} \mathbf{B} = k\mathbf{A}\mathbf{B}$$

15. Based on the rules of matrix multiplication, sums arise from multiplying a row of ones into a column, i.e.

$$(1 \ 1 \ \dots 1) \begin{pmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{pmatrix} = \sum_{i=1}^n a_i$$

Sums of cross products arise from any multiplication of a row with a column, i.e.

$$\begin{pmatrix} b_1 & b_2 & \cdot & \cdot & b_n \end{pmatrix} \begin{pmatrix} a_1 \\ a_2 \\ \cdot \\ \cdot \\ a_n \end{pmatrix} = \sum_{i=1}^n b_i a_i$$